

MADISON SHERIDAN

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Dear Hiring Manager,

I am writing to express my interest in the **POSITION** at **COMPANY**, where I can apply my expertise in computational modeling, quantitative analysis, and algorithmic problem-solving to support data-driven investment and risk management decisions.

My Ph.D. research focused on designing and validating high-fidelity numerical methods for complex, nonlinear systems—experience that translates directly to financial modeling, forecasting, and simulation of dynamic markets. I developed and implemented algorithms for large-scale numerical optimization, uncertainty quantification, and data analysis. I am interested in the financial markets and motivated by the challenge of developing and improving quantitative tools that leverage mathematical intuition to better capture and interpret underlying market trends.

I bring a strong foundation in software development and computational analysis. My work has involved implementing advanced algorithms in C++ and Python, automating workflows for high-performance computing environments, and using MATLAB for statistical analysis and model verification. I am experienced with collaborative coding practices, version control, and model validation, which I see as critical components of modern quantitative research and financial engineering.

Beyond my technical skills, I thrive in interdisciplinary settings where quantitative rigor meets strategic insight. I have collaborated with applied mathematicians, engineers, and scientists to develop reliable, data-driven models and have presented these results to technical and non-technical audiences alike. I am eager to bring this same analytical precision and collaborative approach to a financial institution focused on innovation and quantitative excellence.

I would welcome the opportunity to contribute my modeling background, programming expertise, and problem-solving mindset to your team. Thank you for your time and consideration.

Kind Regards,



Madison Sheridan
Applicant